LAB Assignment 2
Operating Systems (ULC-404)

Instructions: The instructor is required to discuss the following questions and commands with the students. Students are required to make note on these

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1. Introduction of internal and external commands.

2. Feeding output of one command to another command by pipelining.

3. expr, locating command.

4. echo command.

5. Using . and ..

6. Ways for signing off from linux.

7. Ping, Man and help command.

8. Combining the commands.

9. File permissions and changing the access rights (chmod).

10. vi editor and its basics: write a small paragraph using vi editor.

Self study :
1. grep, sort, ps command

2. Study Basics of shell programming.

Shell Programming Basics

1. A shell program is nothing but a series of unix commands.

2. Instead of specifying one job at a time, the shell is given a to-do-list of a program that carries out an entire procedure.

3. Such programs are known as shell scripts.

4. Shell programming language incorporates most of the features that most modern-day programming languages offer.

Shell variables – Rules for building shell variables are as follows:
1. A variable name is any combination of alphabets, digits and an underscore ('_').

2. No commas or blanks are allowed within a variable name.

3. The first character of a variable name must either be an alphabet or an underscore.

4. Variable names should be of any reasonable length.
5. Variable names are case sensitive.

Input/Output

1. Keywords for accepting input – read
2. Displaying output – echo

Assigning value to variables –

1. Values can be assigned to variables through read statement or also by using a simple assignment operator. For ex: age=30

Note : While assigning values to variables using assignment operator, no spaces to be given on either side of it. If the variable doesn't exist it will be created and value assigned

Note : To print or access value of a variable use '$' .

For ex: To print value of variable 'flag' write - echo $flag

Arithmetic in Shell script -

1. All shell variables are string variables, hence to carry out arithmetic operations use expr command which evaluates arithmetic expressions.
2. More than one assignment can be done in a single statement.
3. Before and at the end of expr keyword use ` (back quote) sign not the (single quote i.e.') sign which is generally above TAB key.
4. Terms of the expression provided to expr must be separated by blanks. Thus expression expr 10+20 is invalid.
5. The '*" symbol must be preceded by a \ ,otherwise the shell treats it as a wildcard character for all files in the current directory.

OPERATORS USED IN SHELL SCRIPT – OPERATOR MEANING

1. –gt Greater than
2. –lt Less than
3. –ge Greater than or equal to
4. –le Less than or equal to
5. –ne Not equal to
6. –eq Equal to
7. –a Logical AND

8. –o Logical OR
9. ! Logical NOT

CONTROL INSTRUCTIONS IN SHELLS –

There are four types of control instructions in shell :

1. Sequence Control Instruction.
2. Selection or Decision Control Instruction
3. Repetition or Loop control Instruction
4. Case Control Instruction

Decision statements –

If-then-else-fi statements:

if condition then Commands else Commands fi

For statements:

```
for control variable in value1 value 2 value3 do
Command list done
```

While statements:

```
while control command do
Command list Done
```

Until statements:

```
until control command do
Command list done
```

Case statements:-

```
case value in
    choice 1) commands;;
    choice 2) commands;;
esac
```

Steps to write and execute a script

1. Open the terminal. Go to the directory where you want to create your script.

2. Create a file with . sh extension.

3. Write the script in the file using an editor.

4. Make the script executable with command chmod +x <fileName>.

5. Run the script using ./<fileName>.